Assignment No. 1 -- Linear regression by using Deep Neural network:

Problem Statement: Implement Boston housing price Prediction problem by Linear regression using Deep Neural network. Use Boston House price prediction dataset.

Learning Objectives:

Students should be able to perform Linear regression by using Deep Neural network on Boston House Dataset.

Learning Outcome:

After completing this experiment, students will be able to:

- Understand the concept of Linear regression
- Implement Linear regression by using Deep Neural network
- Analyze and calculate Boston House price

Software Requirements:

Sr No	Software	Specification
1	Operating System	Windows / Linux
2	Programming Language	Python
3	Python Version	Python 3.8 or higher
4	IDE	Jupyter Notebook / Google Colab / VS Code
5	Deep Learning Framework	TensorFlow / Keras
6	Libraries	NumPy, Pandas, Matplotlib, Scikit-learn

Theory:

Contents for Theory:

1. **What is Linear Regression :**
   Linear regression is a statistical approach that is commonly used to model the relationship between a dependent variable and one or more independent variables. It assumes a linear relationship between the variables and uses mathematical methods to estimate the coefficients that best fit the data.

2.Example of Linear Regression

A suitable example of linear regression using deep neural network would be predicting the price of a house based on various features such as the size of the house, the number of bedrooms, the location, and the age of the house.

3.Concept of Deep Neural Network

Deep neural networks are used in a variety of applications, such as image and speech recognition, natural language processing, and recommendation systems. They are capable of learning from vast amounts of data and can automatically extract features from raw data, making them a powerful tool for solving complex problems in a wide range of domains.

4.How Deep Neural Network Work-

Boston House Price Prediction is a common example used to illustrate how a deep neural network can work for regression tasks. The goal of this task is to predict the price of a house in Boston based on various features such as the number of rooms, crime rate, and accessibility to public transportation.

Here's how a deep neural network can work for Boston House Price Prediction:

1. **Data preprocessing:** The first step is to preprocess the data. This involves normalizing the input features to have a mean of 0 and a standard deviation of 1, which helps the network learn more efficiently. The dataset is then split into training and testing sets.

2. **Model architecture:** A deep neural network is then defined with multiple layers. The first layer is the input layer, which takes in the normalized features. This is followed by several hidden layers, which can be deep or shallow. The last layer is the output layer, which predicts the house price.

3. **Model training:** The model is then trained using the training set. During training, the weights and biases of the nodes are adjusted based on the error between the predicted output and the actual output. This is done using an optimization algorithm such as stochastic gradient descent.

4. **Model evaluation:** Once the model is trained, it is evaluated using the testing set. The performance of the model is measured using metrics such as mean squared error or mean absolute error.

5. **Model prediction:** Finally, the trained model can be used to make predictions on new data, such as predicting the price of a new house in Boston based on its features.

6. By using a deep neural network for Boston House Price Prediction, we can obtain accurate predictions based on a large set of input features. This approach is scalable and can be used for other regression tasks as well.

5.Boston House Price Prediction Dataset-

Boston House Price Prediction is a well-known dataset in machine learning and is often used to demonstrate regression analysis techniques. The dataset contains information about 506 houses in Boston, Massachusetts, USA. The goal is to predict the median value of owner-occupied homes in thousands of dollars.

The dataset includes 13 input features, which are:

CRIM: per capita crime rate by town

ZN: proportion of residential land zoned for lots over 25,000 sq.ft.

INDUS: proportion of non-retail business acres per town

CHAS: Charles River dummy variable (1 if tract bounds river; 0 otherwise)

NOX: nitric oxides concentration (parts per 10 million)

RM: average number of rooms per dwelling

AGE: proportion of owner occupied units built prior to 1940 **DIS:** weighted distances to five Boston employment centers.

RAD: index of accessibility to radial highways

TAX: full-valueproperty-taxrateper$10,000 **PTRATIO:** pupil-teacher ratio by town

B: 1000(Bk - 0.63)^2 where Bk is the proportion of black people by town

LSTAT: % lower status of the population

The output variable is the median value of owner-occupied homes in thousands of dollars (MEDV).

To predict the median value of owner-occupied homes, a regression model is trained on the dataset. The model can be a simple linear regression model or a more complex model, such as a deep neural network.

After the model is trained, it can be used to predict the median value of owner-occupied homes based on the input features. The model's accuracy can be evaluated using metrics such as mean squared error or mean absolute error.

Boston House Price Prediction is an example of regression analysis and is often used to teach machine learning concepts. The dataset is also used in research to compare the performance of different regression models.

Conclusion- In this way we can Predict the Boston House Price using Deep Neural Network.

Assignment Question

1.What is Linear Regression?

2.What is a Deep Neural Network?

3.What is the concept of standardization?

4.Why split data into train and test?

5.Write applications of Deep Neural Network?

Assignment No. 2 -- Classification using Deep neural network

Problem Statement:

Binary classification using Deep Neural Networks Example: Classify movie reviews into positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset

Learning Objectives:

Students should be able to classify movie reviews into positive reviews and negative reviews on IMDB Dataset.

Learning Outcome:

After completing this experiment, students will be able to:

Students will be able to automatically classify IMDB movie reviews as positive or negative with measurable accuracy.

Software Requirements:

Sr No	Software	Specification
1	Operating System	Windows / Linux
2	Programming Language	Python
3	Python Version	Python 3.8 or higher
4	IDE	Jupyter Notebook / Google Colab / VS Code
5	Deep Learning Framework	TensorFlow / Keras
6	Libraries	NumPy, Pandas, Matplotlib, Scikit-learn

Theory:

Classification:

Classification is a type of supervised learning in machine learning that involves categorizing data into predefined classes or categories based on a set of features or characteristics. It is used to predict the class of new, unseen data based on the patterns learned from the labelled training data.

In classification, a model is trained on a labelled dataset, where each data point has a known class label. The model learns to associate the input features with the corresponding class labels and can then be used to classify new, unseen data. For example, we can use classification to identify whether an email is spam or not based on its content and metadata, to predict whether a patient has a disease based on their medical records and symptoms, Or to classify images into different categories based on their visual features.

Classification is a common task in deep neural networks, where the goal is to predict the class of an input based on its features. Here's an example of how classification can be performed in a deep neural network using the popular MNIST dataset of handwritten digits.

The MNIST dataset contains 60,000 training images and 10,000 testing images of handwritten digits from 0 to 9. Each image is a grayscale 28x28 pixel image, and the task is to classify each image into one of the 10 classes corresponding to the 10 digits.

We can use a convolutional neural network (CNN) to classify the MNIST dataset. A CNN is a type of deep neural network that is commonly used for image classification tasks.

How Deep Neural Network Work on Classification-

Deep neural networks are commonly used for classification tasks because they can automatically learn to extract relevant features from raw input data and map them to the correct output class.

The basic architecture of a deep neural network for classification consists of three main parts: an input layer, one or more hidden layers, and an output layer. The input layer receives the raw input data, which is usually preprocessed to a fixed size and format. The hidden layers are composed of neurons that apply linear transformations and nonlinear activations to the input features to extract relevant patterns and representations. Finally, the output layer produces the predicted class labels, usually as a probability distribution over the possible classes

IMDB Dataset-The IMDB dataset is a large collection of movie reviews collected from the IMDB website, which is a popular source of user-generated movie ratings and reviews. The dataset consists of50,000 movie reviews, split into 25,000 reviews for training and 25,000 reviews for testing.

Each review is represented as a sequence of words, where each word is represented by an integer index based on its frequency in the dataset. The labels for each review are binary, with 0 indicating a negative review and 1 indicating a positive review.

The IMDB dataset is commonly used as a benchmark for sentiment analysis and text classification tasks, where the goal is to classify the movie reviews as either positive or negative based on their text content. The dataset is challenging because the reviews are often highly subjective and can contain complex language and nuances of meaning, making it difficult for traditional machine learning approaches to accurately classify them. Learning approaches, such as deep neural networks, have achieved state-of-the-art performance on the IMDB dataset by automatically learning to extract relevant features from the raw text data and map them to the correct output class. The IMDB dataset is widely used in research and education for natural language processing and machine learning, as it provides a rich source of labeled text data for training and testing deep learning models.

Conclusion- In this way we can classify the Movie Reviews by using DNN.

<u>Oral Questions:</u>

1.What is Binary Classification?

2.What is binary Cross Entropy?

3.What is Validation Split?

4.What is the Epoch Cycle?

5.What is Adam Optimizer?

Assignment No. 3 -- Convolutional Neural Network (CNN)

Problem Statement: Design and implement a Convolutional Neural Network (CNN) to perform image classification using a standard dataset such as Fashion MNIST or a plant disease dataset.

Learning Objectives:

- To understand the architecture of Convolutional Neural Networks.
- To implement CNN for image classification tasks.
- To analyze model accuracy and performance.
- To gain practical experience with deep learning frameworks.

Learning Outcome:

After completing this experiment, students will be able to:

1. Design CNN architecture for classification problems.
2. Train and evaluate CNN models using real datasets.
3. Interpret performance metrics such as accuracy and loss.

Software and Software Requirements:

Sr No	Software	Specification
1	Operating System	Windows / Linux
2	Programming Language	Python
3	Python Version	Python 3.8 or higher
4	IDE	Jupyter Notebook / Google Colab / VS Code
5	Deep Learning Framework	TensorFlow / Keras
6	Libraries	NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn

Theory:

Convolutional Neural Networks (CNNs) are a class of deep learning models widely used for image processing and computer vision tasks. CNNs automatically learn spatial hierarchies of features from input images using convolution operations.

Key components of CNN include:

1. Convolution Layer

Extracts important features from the input image using filters.

2. Activation Function (ReLU)

Introduces non-linearity into the model.

3. Pooling Layer

Reduces spatial dimensions and computational complexity.

4. Fully Connected Layer

Used for classification tasks.

The **Fashion MNIST dataset** contains grayscale images of clothing items belonging to different categories such as shirts, trousers, shoes, etc.

Algorithm:

1. Import necessary libraries (TensorFlow, Keras, NumPy, Matplotlib).
2. Load the dataset (Fashion MNIST / Plant disease dataset).
3. Preprocess the dataset (normalization and reshaping).
4. Design CNN architecture.
5. Compile the model using optimizer and loss function.
6. Train the model using training data.
7. Evaluate the model using test data.
8. Display classification accuracy.

Result:

The CNN model successfully classified images from the dataset with good accuracy.

Conclusion:

This experiment demonstrated the implementation of a Convolutional Neural Network for image classification. The CNN model effectively extracted features from images and classified them into different categories.

Oral Questions :

1. What is a Convolutional Neural Network?
2. What is the role of pooling layers?
3. What is the difference between CNN and ANN?
4. What is the Fashion MNIST dataset?

Assignment No. 4 -- Recurrent Neural Network (RNN)

Problem Statement: Use the **Google Stock Prices Dataset** and design a **time series analysis and prediction system using Recurrent Neural Network (RNN)**.

Learning Objectives:

- To understand the concept of Recurrent Neural Networks.
- To learn how RNN models handle sequential and time series data.
- To implement stock price prediction using deep learning techniques.
- To analyze trends and patterns in stock market data.

Learning Outcome:

After completing this experiment, students will be able to:

- Understand the concept of time series analysis.
- Implement RNN models for prediction of sequential data.
- Analyze and interpret stock price prediction results.

Software and Software Requirements:

Sr No	Software	Specification
1	Operating System	Windows / Linux
2	Programming Language	Python
3	Python Version	Python 3.8 or higher
4	IDE	Jupyter Notebook / Google Colab / VS Code
5	Deep Learning Framework	TensorFlow / Keras
6	Libraries	NumPy, Pandas, Matplotlib, Scikit-learn

Theory:

A **Recurrent Neural Network (RNN)** is a type of artificial neural network designed for processing **sequential data**. Unlike traditional neural networks, RNNs have feedback connections that allow information to persist over time.

RNNs are particularly useful for **time series analysis**, where past data influences future predictions.

Applications of RNN include:

- Stock price prediction
- Speech recognition
- Natural language processing
- Weather forecasting

The **Google Stock Prices Dataset** contains historical stock market data such as opening price, closing price, high, low, and trading volume. Using RNN, the model learns patterns from past stock prices and predicts future trends.

Algorithm:

1. Import necessary libraries such as NumPy, Pandas, Matplotlib, and TensorFlow/Keras.
2. Load the Google Stock Price dataset.
3. Perform data preprocessing and normalization.
4. Prepare the training and testing datasets.
5. Build the RNN model architecture.
6. Train the model using historical stock price data.
7. Predict stock prices using the trained model.
8. Plot predicted stock prices and compare them with actual prices.

Result:
The RNN model successfully analyzed historical stock price data and predicted future stock price trends with reasonable accuracy.

Conclusion:
This experiment demonstrated the implementation of **Recurrent Neural Network for time series prediction**. The RNN model effectively learned patterns from historical stock price data and generated predictions for future stock trends.

Oral Questions:

1. What is a Recurrent Neural Network (RNN)?
2. What is time series data?
3. What are the applications of RNN?
4. What is the difference between CNN and RNN?
5. What is the vanishing gradient problem?